RESDN: A Novel Metric and Method for Energy Efficient Routing in Software Defined Networks

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***Abstract*—Software-defined networking (SDN) paradigm, with the flexible and logically centralized control, enables dynamically minimizing the network energy consumption by redirecting paths of packets. However, the links and switches are designed to accommodate maximum traffic volume and their power consumption is not traffic proportional. Moreover, there exists a trade-off between energy efficiency and network performance that need to be considered together. Addressing these issues, we propose an energy efficiency metric named Ratio for Energy Saving in SDN (RESDN) that quantifies energy efficiency based on link utility intervals. We provide integer programming formulation and method for maximizing the RESDN of the network. To the best of our knowledge, RESDN approach is novel as it measures how links are profitably utilized in terms of the amount of energy they consume with respect to their utility. We analyze our approach considering various metrics of interest and different types of SDN enabled switches. Experiments show that maximizing the RESDN value improves energy efficiency while maintaining acceptable network performance. In comparison to state-of-the-art utility-based heuristics, RESDN method achieves up to 30% better ratio for energy saving, 14.7 watts per switch power saving, 38% link saving, 2 hops decrease in average path length, and 5% improved traffic proportionality.**

***Index Terms*—Software-defined networks, SDN, energy efficiency, ratio for energy saving, routing, heuristics, traffic proportionality.**

I. Introduction

SOFTWARE Defined Networking (SDN) paradigm is based on the concept of forwarding (data) plane and control plane separation. The control and management of the network is done from a logically centralized controller, while the data plane handles the forwarding functionality. SDN provides the powerful feature of network programmability and has been widely adopted by major companies and network equipment vendors. The ease and flexibility of network control has paved the way to several network applications like load balancing, energy efficiency, dynamic routing, advanced security, and traffic engineering [1]–[7].

Global energy consumption has both environmental and economical issues. 10% of the global energy consumption is due to ICT sector out of which 2% is from network components. By 2020 the total electricity cost of cloud data centers is expected to increase by 63% [8], [9]. However, most of the time network resources are under utilized as compared to the service they render. In data center networks, components are utilized 30% to 40% most of the time [10], [11]. The overall power consumption however remains almost the same for varying amount of traffic volume. A practical solution for a traffic proportional energy consumption problem is to sleep/turn off under-utilized components for low traffic volume. Minimizing the number of network components for low traffic leads to network performance degradation [12]–[18].

Energy saving at the cost of network performance degradation, however, is an undesirable result. The trade-off between energy saving and performance is quite a challenging task. Energy efficient solutions mainly focus on minimizing the number of energy consuming devices whereas network performance optimization solutions mainly focus on keeping the devices work at full capacity regardless of low traffic. The two opposing objectives clearly signifies the need for a holistic solution that maintains the trade-off between efficiency and performance.

In SDN, a naive way for measuring the efficiency of a network is to take the ratio of the power used to the maximum power consumption of the network components. A wide range of energy efficiency metrics like Power Usage Efficiency (PUE) have been proposed and used to measure the efficiency of data centers [19], [20]. The Green Grid and Standard Performance Evaluation Corporation (SPEC) are the most widely known efforts among the many. The metrics proposed, however, are either applicable to data centers only or do not consider the utilities of the network resources, and cannot directly be applied to dynamically changing and programmable networks.

Therefore, addressing the need for a metric to measure the energy efficiency of a network with regard to traffic volume and utility of resources, we propose a novel energy efficiency metric, namely RESDN (Ratio for Energy Saving in SDN), that quantifies energy efficiency based on utility intervals defined by minimum and maximum link utility parameters.[1]

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[1]A very preliminary version of this work was presented in [21].

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The contributions of this work are as follows.

- We propose a novel energy efficiency metric RESDN: the ratio for energy saving in SDN that quantifies energy efficiency based on link utility intervals. To the best of our knowledge, the approach is unique as it measures how links are profitably utilized in terms of the amount of energy they consume with respect to their utility.
- We develop Integer Programming (IP) formulation with the objective of maximizing RESDN and propose heuristics algorithm MaxRESDN for achieving the objective.
- We conduct extensive experiments on Mininet network emulator and POX controller using real network traffic traces and present comparative quantitative analysis of MaxRESDN method. The performance metrics of interest are switch power consumption, RESDN value, percentage of links saved, average path length, throughput, delay, and traffic proportionality.
- We simulate the power consumption of two hardware switches (NEC and Zodiac FX) and one virtual switch (Open vSwitch-OvS) that are OpenFlow enabled, and conduct comprehensive experiments to measure the power consumption of our proposed MaxRESDN heuristic algorithm.
- MaxRESDN heuristic algorithm achieves the highest ratio for energy saving which is up to 30% better than similar state-of-the-art utility-based heuristics algorithms. It also saves up to 38% links and exhibits an average path length closer to the best energy saving heuristics that focuses on performance. MaxRESDN has the highest traffic proportional energy consumption which 3 to 5% better than similar utility-based heuristics. To the best of our knowledge, our approach is the first in maintaining the trade-off between energy efficiency, network performance, and traffic proportionality by optimizing RESDN metric.
- Switch power consumption results show that MaxRESDN exhibits on average up to 14.7 watts, 10 watts, and 3.2 watts less power consumption for NEC, OVS and Zodiac FX switches respectively as compared to other utility-based heuristics for energy efficient routing.
- As the utility parameters directly impact the performance of the MaxRESDN heuristics, we conduct a detailed analysis of the parameters with respect to a range of traffic volumes.

The remainder of the paper is organized as follows. Section II presents related work on energy efficiency in SDN and energy efficiency metrics. Section III presents preliminary work on utility based traffic proportional energy saving in SDN. Section IV describes RESDN metric, provides IP formulation for RESDN maximization, and presents the MaxRESDN heuristics method that maximizes the RESDN of a network. Section V discusses the platform used for our experiments along with heuristics used for comparison and the characteristics of the hardware switches used. Section VI presents our comprehensive experimental results. Section VII concludes and states future directions.

TABLE I
CLASSIFICATION OF ENERGY EFFICIENT ROUTING TECHNIQUES IN SDN

Approach	Topology	Utility Based	Link	Queue Engineering
ElasticTree [12]	Fat-tree	-	✓	-
Carrier Grade [11]	General	-	✓	-
CARPO [13]	Fat-tree	-	✓	-
EnableOpenflow [14]	Fat-tree	-	✓	-
Traffic distribution [32]	General	✓	✓	-
GreenSDN [15]	Bcubic & Fat-tree	-	✓	✓
Fine-grained [33]	General	✓	✓	-
OpenNaas [16]	-	-	✓	✓
Orchestrate [18]	-	-	✓	✓
Utility-based [26]	General	✓	✓	-
Dynamic TA [25]	General	-	✓	-
REsPoNse [23]	-	-	-	-
NetFPGA QE [24]	-	-	-	✓
GreenRE [27]	General	-	✓	-
Bandwidth-aware [28]	Bcubic & Fat-tree	- ✓	-	✓
RA-TAH [31]	Fat-tree	-	✓	✓
TE based [34]	General	-	✓	-
Re-Routing [35]	General	-	-	✓
Resource-aware [36]	Fat-tree	-	✓	-
SDN / Ethernet [37]	General	-	✓	-

II. RELATED WORK

The control and forwarding plane separation in SDN has led to flexibility for many network services ranging from load balancing, dynamic routing, energy efficient and flexible network control [22]. The primary challenge of energy efficiency in networking is the fact that energy consumption is not proportional to the volume of traffic. The problem is even more difficult in the traditional networking since there is a very limited flexibility. Traffic-aware energy efficient routing techniques in SDN attempt to make energy consumption proportional to the traffic volume. We discuss related work in traffic-aware energy efficient routing in Section II-A. Measuring the energy efficiency of a network environment including data centers have been investigated and various metrics were also proposed. In Section II-B, we discuss related work in this area and why we needed a new metrics for SDN.

A. Traffic Aware Energy Efficient Routing Techniques

Traffic aware energy efficient routing techniques in SDN can be classified based on the energy saving capabilities they focus on and the topology structure they are designed for. The main energy saving capabilities are the links and the switches. Some of the works [13]–[16], [18] focus on links, where some others focus on forwarding switches [23], [24]. There also exist works [12], [25], [26] that consider the combination of links and switches as the energy saving components. Furthermore, queue engineering based techniques consider the arrival of packets and their waiting times to decide per-port power requirement [15], [16], [18], [24], [27]–[29].

Based on the topology assumption, energy efficient methods can be classified as general or tailored to a specific network topology. Some methods are general in the sense that they are applicable to any kind of network topology [25], [26], [30] and some are only applicable to specific topologies such as Fat-tree, butterfly, and BCube, where Fat-tree and BCube topologies are among the widely used structures for organizing end systems in data centers [11], [12], [14], [31].

Table I presents the summary of traffic aware energy efficient techniques in SDN. The topology column shows the kind

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of topology structure the approach is designed for and tested on. The utility-based column indicates if the approach focuses on the utility of the links. The link column shows if the method considers links as energy saving component. Queue engineering column shows if the approach applies queue engineering techniques.

Energy efficient techniques utilize the approach of sleeping or turning off unused switches or links when the traffic volume is low and turn them on when the traffic volume increases [14], [28], [38]. The objective of energy efficiency is reducing the number of active network components. However, this has an adverse effect on the performance of the network hence degrades the network performance. There exists a trade-off between the two opposite objectives which are minimizing energy consumption and increasing network performance.

A closely related work considers the remaining bandwidth of links [28]. The problem is formulated using IP, and the proposed heuristics schedules paths for the new flows by considering the paths of the preceding flows. Unlike this approach, our focus is not on the remaining bandwidth of the links but the utilities of the links. We measure the amount each link is utilized based on the utility interval defined by the minimum and maximum utility parameters.

In contrast to the approach of turning off links for energy saving [13], [26], [27], [39], the work [32] examines if turning off/sleeping switches always results in energy saving. The network is modeled as a graph with the nodes as the switches and the edges as the links. Comprehensive experiments show that for the cases where the routing power consumption is cubic with respect to traffic volume, distributing the traffic to underutilized links exhibits better energy saving as compared to turning them off. Since the routing power consumption is polynomial in general, distributing traffic over underutilized links is only applicable to specific cases [32].

Another closely related approach formulates the energy efficiency problem with mixed integer programming (MIP), then proposes heuristic algorithms [25]. There exist four variations of the heuristics where two of them namely Shortest Path First (SPF) and Shortest Path Last (SPL) sort the flows in the order of the shortest path. The other two heuristics namely Smallest Demand First (SDF) and Highest Demand First (HDF) sort the flows according to the flow rate.

In contrast, our approach is flow order independent, it focuses not only on energy efficiency but also on network performance, considers the utility of each link, and proposes a link utility-based energy efficiency metric that simultaneously quantifies efficiency and performance.

B. Energy Efficiency Metrics

Data center energy efficiency metrics have been studied widely with companies and standardization organizations such as the Green Grid, Standard Performance Evaluation Corporation (SPEC), and Transaction Processing Performance Council (TPC) [40], [41]. The Green grid consortium proposes several metrics to measure infrastructure power consumption, carbon dioxide emission, water usage efficiency, and the rate of useful information processed related to resources used. Power Usage Effectiveness (PUE), Data Center Infrastructure Efficiency (DCiE), Computing Power Efficiency (CPE), Carbon Usage Effectiveness (CUE), Water Usage Effectiveness (WUE), and Data Center Productivity (DCP), all of which measure energy efficiency and sustainability. Table II shows the description and formulas of the well known energy efficient metrics in data centers.

TABLE II
ENERGY EFFICIENCY METRICS FOR DATA CENTERS

Metric	Meaning	Formula
PUE	Power Usage Efficiency	$\frac{Total\ DC\ Power}{IT\ Equipment\ Power}$
DCiE	Data Center Infrastructure Efficiency	$\frac{1}{PUE}$
CPE	Computing Power Efficiency	$\frac{IT\ equipment\ Utilization}{PUE}$
CUE	Carbon Usage Efficiency	$PUE \cdot Electricity\ Carbon\ Emission\ Rate$
WUE	Water Usage Effectiveness	$\frac{Anual\ DC\ water\ usage}{IT\ equipment\ energy\ consumption}$
DCP	Data Center Productivity	$\frac{Useful\ work\ amount\ produced}{Resource\ for\ producing\ the\ work}$

Power Usage Effectiveness (PUE), the most prevalent metric, is defined as the ratio of the total annual amount of energy that comes into a data center to the energy consumed by the IT equipment [42]. IT equipment includes workstations, storage, monitors, and network devices. Total data center energy consists of IT equipment energy consumption plus energy consumed for cooling system components, UPS, switch gear, and data center lightening. The ideal value for PUE is 1 which means all the energy in the data center is used by the IT equipment. The main limitations of PUE metric are that it is only applicable to a single building that supports a data center [39], and it does not focus specifically on links and switches which are the main energy saving capabilities in SDN.

Data Center Infrastructure Efficiency (DCiE), which is the inverse of PUE, is calculated as the ratio of IT equipment power consumption to total the facility power consumption. Carbon Usage Effectiveness (CUE) is the product of the amount of Carbon Emission Factor (CEF) and the data center's annual PUE [43]. CEF is the ratio of carbon dioxide emitted in kg per unit of energy in kWh. Computing Power Efficiency (CPE) scales the IT equipment utilization from 0 to 1, and is calculated as the ratio of IT equipment utilization to PUE. Water Usage Effectiveness (WUE) is the ratio of the annual site water usage in liters to the IT equipment energy usage in kilowatt hours (Kwh) [44]. WUE measures the water consumption in relation to IT equipment energy consumption. However, it is only applicable to a single data center site and does not consider network devices which are the main focus of energy saving in SDN. Data Center Productivity (DCP) measures the amount of useful information processing completed relative to the amount of resources consumed in producing the task [19]. Since the task can use several resources which are measured with different units, quantifying their utility with a single value is not straightforward.

To the best of our knowledge, there is no metric that fully focuses on network components such as switches and links which are the energy saving capabilities in SDN. Using the existing metrics in SDN environment falls short of capturing

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the link utility with regard to traffic as an important energy efficiency parameter. Another drawback of the metrics is that they are measured annually and offline. Therefore, the need for a new metric that can adapt in response to the change in traffic and network environment is of great significance. Failing to do so undermines the flexible network management capability that could have been exploited by SDN.

III. Preliminaries

Next Shortest Path (NSP) and Next Maximum Utility (NMU) heuristics were proposed in our prior work [26] that are generally applicable to any network topology and are flow order independent. NSP and NMU first select those links with utility less than *Umin* as the candidate links, then aim at redirecting flows passing through the candidate links to the next shortest path or the path in the direction of the link with maximum utility, respectively. The rationale behind selecting links with utilities less than *Umin* as candidate is to redirect flows that pass through these links to make them inactive. NSP and NMU attempt to put underutilized links to inactive state. Doing so not only reduces the number of active links, but also increases the overall utilities of the links. Next, we discuss the details of NSP algorithm which attempts to make every link to be utilized within the minimum utility *Umin* and maximum utility *Umin* values or to be made inactive.

NSP (Algorithm 1) identifies under-utilized links and reroutes traffic flows passing through them to the next shortest alternative path. The algorithm takes topology as a graph $\mathbb{G}$, set of flows $\mathbb{F}$, values *Umin*, and *Umax* as inputs. It first initializes the link utilities by taking the current utilities of the links, and then selects the under-utilized links and sets the *CandidateList*. Line 6 returns the list of paths from Z_i to Z_j excluding those links in the *CandidateList*. The paths stored in $shortestpaths_{ij}$ are sorted according to the path length. The $shortestpaths_{ij}$ is set to $Path_{ij}$ (line 7). NSP algorithm then picks one of the shortest paths *SP* to replace the direct link. *SP* is replaced with the next $Path_{ij}$ until all the links in $\forall e_{ab} \in SP$, $U_{ab} + U_{ij} \leq Umax$ (lines 8 to 10). Replacing a particular link with an alternative path needs redirecting all the flows passing through that link. Basically, for each redirected flow, the utilities of the links in *SP* are increased and the utility of the direct link is decreased (lines 13 and15). The status of a link is updated if its utility is 0 (i.e., all flows passing through are redirected to alternative path *SP*), hence the graph $\mathbb{G}$'s status is updated (lines 17-21). Algorithm 1 returns the subgraph of $\mathbb{G}$.

Algorithm 1 NSP: NextShortestPath($\mathbb{G}$, $\mathbb{F}$, *Umin*, *Umax*)

1: **Input:** Graph $\mathbb{G}$, set of traffic flow $\mathbb{F}$, minimum utility *Umin*, and maximum utility *Umax*
2: **Output:** Subgraph of $\mathbb{G}$
3: $\mathbb{U} \leftarrow NetworkStatus(\mathbb{G}, \mathbb{F})$
4: $CandidateList = ReturnCandidateList(\mathbb{U}, Umin)$
5: **for all** $e_{ij} \in CandidateList$ **do**
6: $\quad Path_{ij} \leftarrow shortestpaths_{ij}$
7: $\quad SP = Path_{ij}[0]$ ▷ Pick one of the shortest paths
8: $\quad$ **while** $\exists\ e_{ab} \in SP$ where $U_{ab} + U_{ij} > Umax$ **do**
9: $\quad\quad SP = \texttt{next}(Path_{ij})$
10: $\quad$ **end while**
11: $\quad$ **for all** f `passing through` e_{ij} **do**
12: $\quad\quad$ **for all** $e_{ab} \in SP$ **do**
13: $\quad\quad\quad U_{ab} = U_{ab} + \frac{\lambda_f}{W_{ab}}$ ▷ increment U_{ab}
14: $\quad\quad$ **end for**
15: $\quad\quad U_{ij} = U_{ij} - \frac{\lambda_f}{W_{ij}}$ ▷ decrement U_{ij}
16: $\quad$ **end for**
17: $\quad$ **if** $U_{ij} == 0$ **then**
18: $\quad\quad L_{ij} = 0$ ▷ state of link is inactive
19: $\quad\quad$ `Turn off(`e_{ij}
20: $\quad\quad$ `Update G`
21: $\quad$ **end if**
22: **end for**

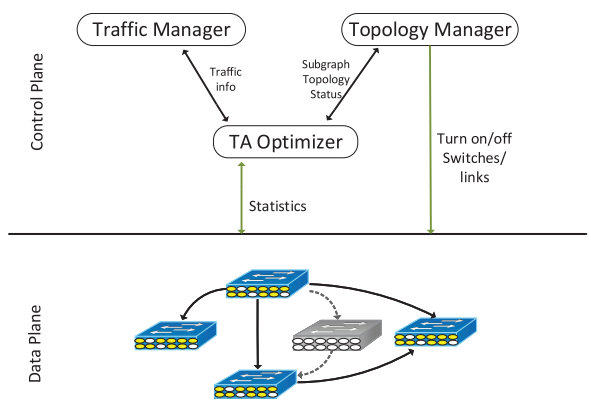

Fig. 1. The framework for traffic proportional energy efficiency in SDN.

IV. RESDN Metric: Ratio for Energy Saving

In this section, we present our traffic proportional framework for energy efficiency in SDN, and propose RESDN energy efficiency metric for SDN, based on link utility interval defined by the minimum and maximum utility parameters *Umin* and *Umax*. Then, we present IP formulation and heuristics to maximize the RESDN value of a software defined network in a dynamic environment.

A. Traffic Proportional Energy Efficiency Framework

As illustrated in Figure 1, our traffic proportional energy efficiency framework for SDN consists of three components, namely Traffic Manager, Topology Manager, and Traffic-Aware (TA) Optimizer.

The information of the network traffic generated by the applications are passed to the traffic manager. Then, the traffic manager stores and passes this traffic information to the TA optimizer. The statistics of the network and the topology information is fed to the TA optimizer from the switches. The optimizer generates an optimal subgraph based on the traffic volume and the utility of the links. Since the framework is designed to work in a dynamic environment, a low traffic load should result in a subgraph with smaller number of active links and switches, whereas high traffic load should increase the number of active links and switches in the subgraph respectively. In the former, the sleep/turn-off decision for a subset links and switches would be achieved. In the latter, however, a subset of links and switches would be turned on or made active [45].

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TABLE III
PARAMETERS AND THEIR DESCRIPTION

Parameter	Description
$\mathbb{G}$	Network graph that consists of $\mathbb{Z}$ and $\mathbb{E}$
$\mathbb{Z}$	Set of switches $Z_i \in \mathbb{Z}$ represents switch i
S_i	Binary variable of switch Z_i
$\mathbb{E}$	Set of links $e_{ij} \in \mathbb{E}$ is a link between switches Z_i and Z_j
L_{ij}	Binary variable of link e_{ij} (active or inactive)
W_{ij}	Bandwidth of the link e_{ij}
$\mathbb{F}$	Set of flows $f \in \mathbb{F} = (sr, ds, \lambda_f)$
f	A flow f=(sr, ds, λ_f) with source , destination and flow rate
F_{ij}	Binary variable of flow f passing through link e_{ij}
$\mathbb{U}$	Set of utilities of all links
U_{ij}	Utility of link e_{ij}
X_{ij}	Binary variable of link e_{ij} $Umin \leq U_{ij} \leq Umax$

B. Motivation and Definition for RESDN Metric

Traffic aware energy saving approaches attempt to turn off/sleep network components by observing network traffic volume. Traffic proportional energy consumption is more strict requirement than the traffic awareness. It demands the percentage of power consumption of network components to be proportional to the traffic volume. However, none of the existing energy efficiency metrics discussed in Section II take the utilities of the network components under consideration with regard to the network traffic volume. Addressing this issue, we propose Ratio for Energy Saving in SDN (RESDN) metric that indicates the percentage of links with utilities within the interval defined by *Umin* and *Umax* with respect to the total number of active links.

The RESDN value reflects how the network components are profitably utilized. As an analogy, the current mode of payment for customers in cloud computing is Pay-as-you-go (PAYG) which charges based on the usage of computing resources. Similarly, RESDN enables the network provider to implement Pay-as-you-use (PAYU) in SDN for energy consumption. A simple scenario, in this case, is a network provider aiming to pay the energy cost of a network component only if it is utilized within a utility interval expressed in terms of RESDN. This RESDN policy is defined by *Umin* and *Umax* parameters and enforced by the controller. The controller decides the links and their corresponding utilities to match the utility interval specified by the provider. The technical solution to this is to redirect the flows passing through underutilized links to other links. Especially in a dynamic network environment where the traffic volume increases and decreases, the RESDN value adapts in response to the change. The controller is responsible for keeping the RESDN value at maximum. Therefore, the energy cost of the network provider would be proportional to the traffic volume being streamed in the network at any time.

Table III shows the parameters used in the formulation of the RESDN. The network is modeled as a weighted graph $\mathbb{G} = (\mathbb{Z}, \mathbb{E})$ with $\mathbb{Z}$ as the set of switches where $Z_i \in \mathbb{Z}$ represents switch *i* and $e_{ij} \in \mathbb{E}$ represents that there exists a link between switches Z_i and Z_j. The weight W_{ij} corresponds to the bandwidth of the link e_{ij}.

Let binary variable S_i denote the status of switch Z_i such that

$$S_i = \begin{cases} 1, & \text{if switch } Z_i \text{ is active} \\ 0, & \text{otherwise} \end{cases}$$

Network traffic is represented by the set of flows $\mathbb{F}$ where $f \in \mathbb{F}$ is defined as $f = (sr, ds, \lambda_f)$. sr and $ds \in \mathbb{Z}$ are the source and destination switches of flow *f*, and λ_f is the rate of f measured in bits per second.

$$F_{ij} = \begin{cases} 1, & \text{if flow } f \text{ passes through edge } e_{ij} \\ 0, & \text{otherwise} \end{cases}$$

$\mathbb{U}$ is the set of the utilities of every edge in the graph $\mathbb{G}$ where $U_{ij} \in \mathbb{U}$ is defined as the ratio of the sum of the rates of the flows passing through the edge e_{ij} to the link bandwidth W_{ij}.

$$U_{ij} = \frac{\sum_{\forall f} F_{ij} x \lambda_f}{W_{ij}} \tag{1}$$

Utility value of a link is between 0 and 1, where 0 means no flow is passing through the link and 1 means the sum of the flow rates passing through the link is equal to the link bandwidth. Let *Umin* be the minimum utility value to keep a link active and *Umax* is the maximum link utility.

$$X_{ij} = \begin{cases} 1, & Umin \leq U_{ij} \leq Umax \\ 0, & \text{otherwise} \end{cases}$$

The energy efficiency metric is formally defined as

$$RESDN = \frac{\sum_{\forall e_{ij}} X_{ij}}{\sum_{\forall e_{ij}} L_{ij}} \tag{2}$$

where L_{ij} is binary variable that denotes the status of edge e_{ij}

$$L_{ij} = \begin{cases} 1, & \text{if edge } e_{ij} \text{ is active} \\ 0, & \text{otherwise} \end{cases}$$

According to Equation (2), RESDN value of 1 means that all links that are turned on are operating between the interval defined by parameters *Umin* and *Umax*. RESDN value of 0 means that none of the links are operating profitably which means that the network is underutilized or over-utilized. RESDN values close to 1 indicate that the majority of the links are operating within the utility interval (i.e., between *Umin* and *Umax* values) meaning both high energy efficiency and acceptable network performance is achieved. On the other hand, RESDN values close to 0 indicate either poor energy efficiency or poor network performance since it shows that most of the utilities of the links are below *Umin* or above *Umax*.

The motivation behind the RESDN is based on the idea that network providers would like to pay for energy consumption for a given resource only if it is utilized at least to a minimum value of *Umin*. If the utility of a resource is less than *Umin*, then it is underutilized. *Umax* needs to be less than 1 since setting it to 1 would create network congestion which would degrade network performance. The larger the value of RESDN, the more the percentage of the active links that operate between the *Umin* and *Umax* values. Smaller RESDN value shows that a large percentage of active links operate unprofitably.

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C. IP Formulation for RESDN Optimization

The objective function of our IP model is to maximize the RESDN value of the network.

$$\text{maximize} \quad RESDN \tag{3}$$

$$\text{subject to} \quad \sum_{\forall f} F_{ij} x \lambda_f \leq W_{ij}, \ \forall e_{ij} \tag{4}$$

$$\sum_{\forall f} F_{ai} = \sum_{\forall f} F_{ib} \ , Z_i \neq f^{sr}, Z_i \neq f^{ds} \tag{5}$$

$$F_{mj} = F_{in}, \ \forall f \ \ Z_m = f^{sr}, \ \ Z_n = f^{ds}, \ \forall e_{mj}, \ \exists e_{in} \tag{6}$$

$$F_{ij} \leq S_j \ \ and \ \ F_{ji} \leq S_j, \ \forall Z_j \in \mathbb{Z} \tag{7}$$

$$S_i \leq \sum_{\forall f} \left[F_{ij} + F_{ji} \right], \ \forall Z_i \in \mathbb{Z} \tag{8}$$

$$L_{ij} \leq S_i \text{ and } L_{ij} \leq S_j \ \forall Z_i, Z_j \in \mathbb{Z} \tag{9}$$

The constraints in the formulation are as follows. Equation (4) states that the sum of the rates of flows between two switches should not exceed the corresponding link capacity. Equation (5) states that the number of flows entering and leaving switches which are neither destination nor sources of flow should be equal. Equation (6) assures a flow entering from source switch should reach the destination switch. Equation (7) asserts that a flow should not be assigned to a link that is connected to an inactive switch. Equation (8) models the relationship between the flows passing through a link and a switch. It asserts that if no flow is passing through all the links connected to switch Z_i, then change the binary variable $\mathrm{S}_i = 0$. Equation (9) asserts that a link should be put in active state if and only if both of the switches it is connecting are active, otherwise, it is inactive.

D. MaxRESDN Heuristics Method

The energy efficient routing is an NP-hard problem [12], [13], [37]. Formal solutions can best fit for a small number of network switches but fail to scale up when the number is in the order of hundreds. However, in cloud data centers where the number of physical machines is in the order of thousand, the number of switches is in the order of hundreds. That triggers the need for heuristics algorithms that can run in reasonable polynomial time to provide sub-optimal solutions.

We propose the heuristics algorithm named MaxRESDN (Maximize RESDN). Algorithm 2 aims to maximize the RESDN value of the network as it assigns a path to each flow. The inputs for the MaxRESDN are the network represented as graph $\mathbb{G}$, the current utility of links $\mathbb{U}$, set of flows $\mathbb{F}$, the minimum utility *Umin* and maximum utility *Umax*. Each flow $f = (sr, ds, \lambda_f) \in \mathbb{F}$ is expressed as source *sr*, destination *ds*, and flow rate λ_f which correspond to source address, destination address, and rate of f measured in bits per second (line 3). Line 6 increments the utilities of all the links on the path assigned to flow f by the $\frac{\lambda_f}{W_{ij}}$ factor. Line 4 assigns each flow to the path that maximizes the RESDN among alternatives using the *PathMaxRESDN* $(\mathrm{sr}, \mathrm{ds}, \lambda_f)$ method. Lines 9 to 13 makes link e_{ij} inactive if its corresponding utility U_{ij} is 0.

Algorithm 2 MaxRESDN ($\mathbb{G}$, $\mathbb{F}$, $\mathbb{U}$, *Umin*, *Umax*)

1: **Input:** Graph $\mathbb{G}$, set of traffic flow $\mathbb{F}$, utility of links $\mathbb{U}$, minimum utility *Umin*, and maximum utility *Umax*
2: **Output:** Modified utility of links $\mathbb{U}$ and graph $\mathbb{G}$
3: **for all** $f = (sr, ds, \lambda_f) \in F$ **do**
4: $\quad path_f \leftarrow \mathrm{PathMaxRESDN} \ (\mathrm{sr}, \mathrm{ds}, \lambda_f)$
5: $\quad$ **for all** $e_{ij} \in path_f$ **do**
6: $\quad\quad U_{ij} \leftarrow U_{ij} + \frac{\lambda_f}{W_{ij}}$
7: $\quad$ **end for**
8: **end for**
9: **for all** $e_{ij} \in \mathbb{E}$ **do**
10: $\quad$ **if** $U_{ij} == 0$ **then**
11: $\quad\quad L_{ij} \leftarrow 0$
12: $\quad$ **end if**
13: **end for**

Algorithm 3 PathMaxRESDN(sr, ds, λ_f)

1: **Input:** Graph $\mathbb{G}$,utility of links $\mathbb{U}$, minimum utility *Umin*, maximum utility *Umax*, source node sr, destination node ds, and flow rate λ_f
2: **Output:** path_f for flow $\mathrm{f}(sr, ds, \lambda_f)$ that maximizes RESDN
3: $AllPath \leftarrow$ all paths between sr and ds
4: **for all** `Path` $\mathrm{P} \in AllPath$ **do**
5: $\quad$ **for all** links $e_{ab} \in \mathrm{P}$ **do**
6: $\quad\quad$ **if** $U_{ab} + \lambda_f > UMax$ **then**
7: $\quad\quad\quad$ *AllPath*.remove(P)
8: $\quad\quad$ **end if**
9: $\quad$ **end for**
10: **end for**
11: $\mathrm{Path}_f \leftarrow MaxRESDN(Allpath)$

Algorithm 3 (PathMaxRESDN) returns the path that maximizes RESDN for a given flow f with source sr, destination ds, and flow rate λ_f. Line 3 initializes the *Allpath* list with all paths from sr to ds of flow f. Lines from 5 to 9, ensure the stability of the network by making the utilities of all links in each path plus the rate of flow of f not to exceed the *Umax* value. While line 11 selects the path with maximum RESDN value and assign it to Path_f.

The time complexity of the MaxRESDN algorithm depends on the total number of flows $|\mathbb{F}|$ and the PathMaxRESDN (Algorithm 3). Since the controller has a global view of the network, for each source destination switch pair in the network, all the paths from source to destination are stored in descending order with their RESDN value. The path with the maximum RESDN value then can be extracted in constant time. Therefore, the complexity of the MaxRESDN algorithm is $\mathrm{O}(|\mathbb{F}|)$.

E. Practical Use Cases

A practical use case is in the context of Network as a Service (NaaS), a type of Infrastructure as a Service (IaaS), where the infrastructure is the network itself. Our RESDN metric helps the service provider to proactively set the link utility

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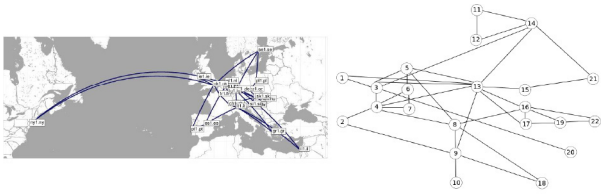

(a) (b)

Fig. 2. GEANT network topology (a) Map view, (b) Graph view.

interval parameters which refer to its profit margin. Then by maximizing the RESDN value of the network, our framework will maintain the trade-off between performance and efficiency. The network service provider may present services with different RESDN, considering RESDN as part of SLA with its users.

As another use case, efficiency and performance metrics for data centers can be considered. Since devices are designed to accommodate the peak load and work with full capacity regardless the change in traffic volume, most devices are under-utilized. Forcing links to work within link utility interval parameters maintains the trade-off between efficiency and performance. The RESDN metric is designed to measure the energy efficiency and performance of a data center network in a scale of 0 to 1.

V. Experimental Platform

This section presents the platform used to conduct our experiments. First, the simulation setup and the performance metrics are discussed, and then utility based heuristics algorithms used for comparison are presented. Finally, the characteristics of three OpenFlow protocol enabled switches used in our experiments for measuring power consumption are presented.

A. Setup and Performance Metrics

The experimental platform is constructed using POX controller and Mininet [46] network emulator installed on Ubuntu 16.04 64-bit. The network topology is created on Mininet, and the heuristics are implemented on POX controller. Our experiments are conducted using real traces from SNDlib [47], in particular the GEANT dynamic traffic trace of the European research network.

Figure 2 shows the GEANT topology used in our experiments [47]. The number of nodes and bidirectional links are 22 and 36, respectively. Traffic demand matrices are aggregated for 15 minutes from a 4-month duration trace. There exist 11460 traffic demand matrices in total. The number of flows in each traffic demand matrix ranges from 82 to 462. In order to demonstrate traffic awareness in our experiments, different traffic volumes ranging from 20% to 90% with respect to the network capacity are used.

Table IV presents the performance metrics of interest measured in our experiments, namely average power consumption of switches, RESDN value, percentage of links saved, average path length, throughput, delay, and traffic proportionality.

TABLE IV
Performance Metrics and Measurement Units

Metric	Unit / Description
Average Power Consumption of Switches	Watt
RESDN	metric as percentage
Links saved	percentage of links saved
Average path length	#hops
Throughput	Mbits/sec
Delay	milliseconds
Traffic Proportionality	[0,1]

The network performance is analyzed in terms of average path length, throughput and delay.

Equation (10) shows the total power consumption of a switch (P_{switch}) as the sum of P_{base}, P_{config} and $\mathrm{P}_{control}$ [48]. P_{base} is the power consumption for keeping the switch on without any active ports.

$$P_{switch} = P_{base} + P_{config} + P_{control} \tag{10}$$

The configuration power consumption P_{config} is calculated as

$$P_{config} = \sum_{i}^{N_{activePorts}} c_i.P_{port} \tag{11}$$

where c_i is the percentage of the line speed of the port relative to the maximum line speed, and P_{port} is the power consumption of a port at full capacity measured in watt.

Equation (12) shows the power consumption of control $P_{control}$ where $r_{PacketIn}$ and $E_{PacketIn}$ are the rate and energy consumption of *PacketIn* respectively. Likewise, $r_{FlowMod}$ and $E_{FlowMod}$ are the rate and energy consumption of the *FlowMod* operations.

$$P_{control} = r_{PacketIn} x E_{PacketIn} + r_{FlowMod} x E_{FlowMod} \tag{12}$$

The ratio of energy saving metric is calculated as

$$RESDN = \frac{\sum_{\forall e_{ij}} X_{ij}}{\sum_{\forall e_{ij}} L_{ij}}$$

where L_{ij} is binary variable that denotes the status of edge e_{ij} and X_{ij} is a binary variable that indicates if the utility of the link is between *Umin* and *Umax*. Although, our primary goal is energy saving, RESDN also considers network performance by the *Umax* value.

Links are one of the energy-saving capabilities in a network environment. Minimizing the number of links hence helps reducing the power consumption for keeping a link active and turning off the ports and the switches the link is connecting. The percentage of links saved is calculated as

$$\text{Links saved} = 100\left(1 - \frac{\sum_{\forall e_{ij}} L_{ij}}{|\mathbb{E}|}\right) \tag{13}$$

Theoretically, the average number of hops a flow traverses to reach its destination is directly proportional to the time it takes to reach its destination. The average path length in terms of average number of hops is calculated as

$$\text{Average path length} = \frac{\sum_{\forall f} \sum_{\forall e_{ij}} (F_{ij})}{|\mathbb{F}|} \tag{14}$$

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TABLE V
TABLE OF HEURISTICS

Abbreviation	Description
SPF [25]	Shortest Path First
SPL [25]	Shortest Path Last
SDF [25]	Smallest Demand First
HDF [25]	Highest Demand First
NSP [26]	Next Shortest Path
NMU [26]	Next Maximum Utility
B [26]	Best Combination of NSP (NMU) with others
MaxRESDN	Maximize RESDN

TABLE VI
PARAMETERS OF *Umin* AND *Umax* USED

Traffic volume percentage	20	30	40	50	60	70	80	90
$Umin$	31	28	30	25	20	19	15	12
$Umax$	82	85	90	90	92	95	95	95

TABLE VII
SWITCH POWER CONSUMPTION PARAMETERS

Parameters	NEC PF 5240 [48], [49]	OvS [48]	Zodiac FX [51]
Base[W]	118.33	48.7397	15
P_{port}[W]	0.52	Na	0.15
$E_{PacketIn}$ [μ W/packet]	711.30	775.53	775.53
$E_{FlowMod}$ [μ W/packet]	29.25	356.743	1455.13

Throughput is calculated as the amount of data transferred per unit of time. In Mininet, we use iperf command to measure the throughput of source and destination pairs. In the experiments, we measure the average throughput of all source and destination pairs.

Delay is calculated as the amount of time needed for data packets to be transferred from their source to destination. In our experiments, network delay is measured as the time it takes for the packets of a flow to start at the source node and reach at the destination node. The delay is measured both on the Mininet side and POX controller.

Traffic proportionality is defined as the ratio of the traffic volume percentage to the network power consumption, and is computed as

$$\left(\frac{\%\text{Traffic Volume}}{M\dfrac{\sum_{\forall Z_i} SP_i}{|\mathbb{Z}|} + N\dfrac{\sum_{\forall e_{ij}} L_{ij}}{|\mathbb{E}|}}\right)(M+N) \tag{15}$$

where M:N is the ratio of the power consumption of the switches to the links. In these experiments, the ratio M:N is used as 3:1 [26].

B. Algorithms Used for Comparison

The heuristics algorithms we used for comparison with MaxRESDN heuristics are listed in Table V. The methodology in [25] formulates the problem with MIP and proposes four heuristic algorithms, namely Shortest Path First (SPF), Shortest Path Last (SPL), Smallest Demand First (SDF), and Highest Demand First (HDF). The first flow is assigned to its corresponding shortest path, and the succeeding flows are assigned to paths where the change in energy consumption is minimized. Each heuristics algorithm uses a different criteria (i.e., according to the shortest path first, shortest path last, smallest demand first, and highest demand first) to sort the flows and processes them in the corresponding order.

As explained in Section III, the objective of NSP algorithm is to re-route flows passing through the under-utilized links to the next shortest path. NMU, on the other hand, chooses the path that has the link with maximum utility. While NSP gives priority to performance, NMU focuses on maximizing the utility of active links. Both NSP and NMU are not only flow order independent, but can also be applied on top of other algorithm outputs to improve efficiency. The B heuristics is the application of NSP or NMU algorithms on top of the results of SPF, SPL, SDF, or HDF algorithms. It refers to the best outcome (B) in terms of energy efficiency.

Table VI shows the values of *Umin* and *Umax* parameters used in the experiments for different percentage of traffic volumes. The values that maximize the number of links saved are determined by a greedy-based approach. A detailed analysis of these parameters is presented in Section VI-D.

C. Types of Switches

Table VII shows the parameters used in the power calculation for two hardware switches (NEC PF 5240 and Zodiac FX) and one virtual switch Open vSwitch (OvS) which are OpenFlow enabled. NEC PF 5240 is a hybrid switch technology that adds OpenFlow protocol to the traditional network functionality [48], [49]. Open vSwitch (OvS) is a multilayer virtual switch designed for enabling network automation by supporting various protocols including but not limited to OpenFlow [48], [50]. Zodiac FX is an OpenFlow switch designed for small scale uses [51].

VI. EXPERIMENTAL RESULTS

This section presents our extensive experimental results in terms of switch power consumption, RESDN and energy efficiency, network performance, and analysis of the link utility interval parameters.

A. Switch Power Consumption Results

Figure 3 shows the average power consumption of the switches for the NEC PF 5240 OpenFlow-based switch versus traffic volume. As compared to the best combination (B) of NSP or NMU with other algorithms (Table V), MaxRESDN power consumption is on average 6 to 9 watts less than the B algorithm. As compared to NSP and NMU [26], MaxRESDN has shown up to 14.7 and 10.7 watts less in power consumption, respectively. This indicates that through maximizing the RESDN parameter, improvements in energy savings are achieved for all traffic volumes.

Figure 4 shows the power consumption of Zodiac FX switch for NSP, NMU, B, and MaxRESDN algorithms versus traffic volume. Similar to the previous experiments, the average power consumption of switches is proportional to the traffic volume. However, power consumption of MaxRESDN is on average 3.2, 2.3, and 2 watts less than NSP, NMU and B algorithms, respectively.

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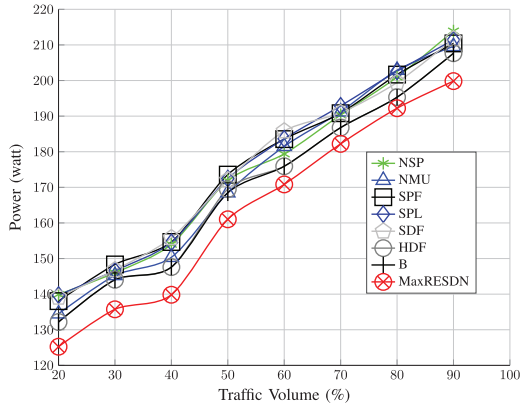

Fig. 3. Power Profile for NEC PF 5240 switch in watt vs a range of traffic volume from 20% to 90%.

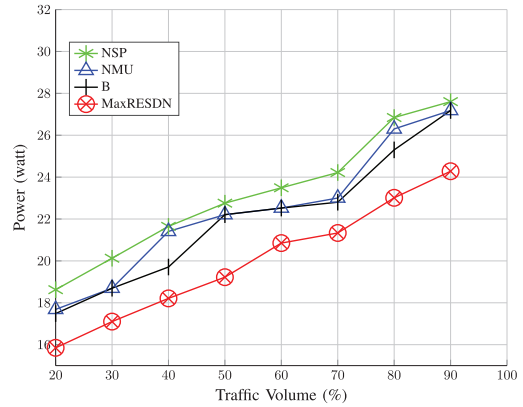

Fig. 4. Power Profile for Zodiac FX switch in watt vs a range of traffic volume from 20% to 90%.

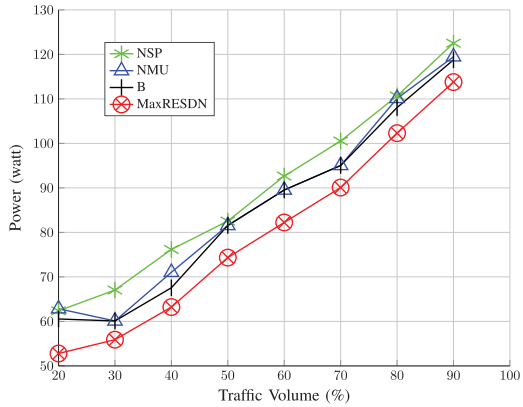

Fig. 5. Power Profile for OvS switch in watt vs a range of traffic volume from 20% to 90%.

Figure 5 shows the power consumption of OvS switches versus traffic volume. Results show that as the traffic volume increases, the average power consumption of switches also increases. The MaxRESDN algorithm exhibits 5.8, 10, and 8.9 watts less energy consumption as compared to B, NSP, and NMU respectively.

Overall, Figures 3, 4, and 5 show that MaxRESDN is the most energy efficient as compared to the other algorithms (Table V). Our findings indicate that maximizing the RESDN

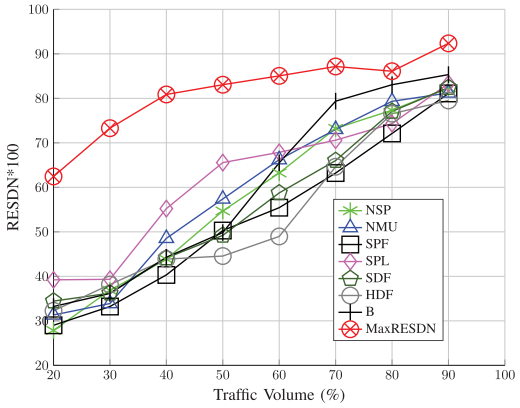

Fig. 6. RESDN Values NSP,NMU, SPF, SPL, SDF, HDF, B and MaxRESDN heuristic algorithms versus a traffic volume ranging from 20% to 90%.

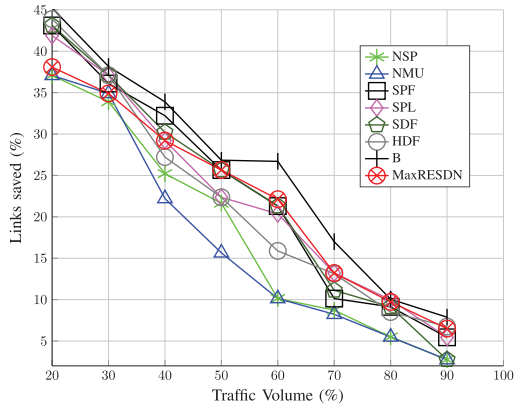

Fig. 7. The percentage of links saved versus traffic volume ranging from 20% to 90%.

value in the network helps minimizing the energy consumption. This is mainly because enforcing links to operate within the minimum and maximum utility parameters minimizes the number of active ports and switches.

B. RESDN and Energy Efficiency Results

Figure 6 shows the RESDN values of all heuristic algorithms as the traffic volume increases. The MaxRESDN algorithm achieves the highest RESDN value which is 22% better in low traffic volumes and 10% better in high traffic volumes. This is because unlike the other algorithms whose objective is to minimize the number of links used in the network, MaxRESDN algorithm aims at maximizing the RESDN value, which increases the utility. In particular, for low traffic volumes, it is more likely for flows that pass through over-utilized links to be redirected to underutilized links.

Figure 7 shows the energy saving in terms of the percentage of links saved. The B heuristics which is the best combination of the NSP or NMU algorithm exhibits the highest energy saving, while NSP and NMU result in the lowest energy saving. MaxRESDN heuristics saves 38% of links for 20% traffic volume, and 6.5% of links for 90% traffic volume.

Figure 8 presents the power consumption of all algorithms for the Zodiac switch. Switch power consumption by MaxRESDN algorithm is 16 to 22 watts which is on average 4

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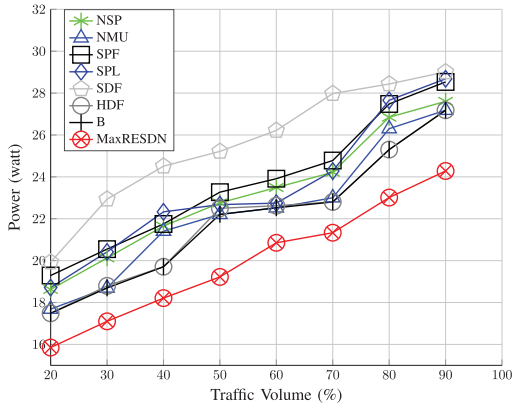

Fig. 8. The average power consumption of the Zodiac switch measured in watts versus traffic volume ranging from 20% to 90%.

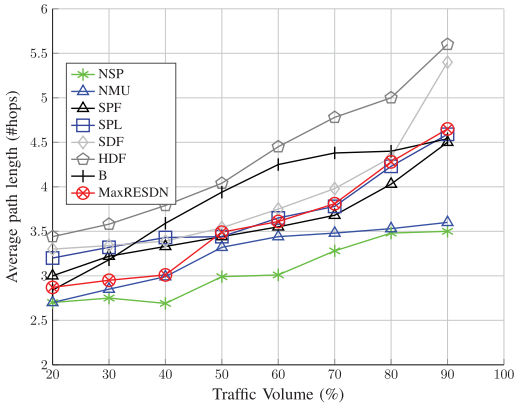

Fig. 9. Average path length in terms of number of hops versus traffic volume ranging from 20% to 90%.

watts less than the SDF algorithm. MaxRESDN also demonstrated a power consumption on average 2 watts less than the B algorithm. These results show that maximizing the RESDN value which is based on link utilities helps reducing the power consumption of switches.

C. Network Performance Results

Figure 9 shows the average path length in terms of the number of hops versus the traffic volume, and is calculated as given in equation (14). The NSP heuristics achieves the best path length since its objective is redirecting flows to the next alternative shortest path. The decisions are made periodically by selecting the underutilized links as candidates and then redirecting the flows. B heuristics which is the combination of NSP and the best of SPF, SPL, SDF, and HDF in terms of energy saving results in improved average path length. This indicates that our previous heuristics can be applied on top of other algorithms and improve both performance and energy saving. The MaxRESDN heuristics performance is closer to NMU that has the objective of maximizing the utilities of the links, by directing the flows passing through underutilized links to more utilized links.

The experimental results show that NSP and NMU have the lowest average path length but the least energy saving, since they first consider network performance rather than energy

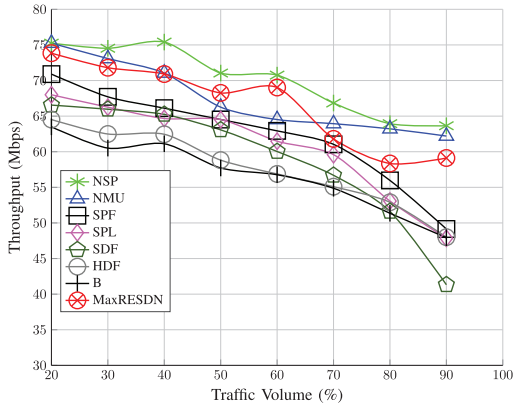

Fig. 10. Throughput in Mbps versus traffic volume ranging from 20% to 90%.

efficiency. The other algorithms in [25] exhibit better energy saving than NSP and NMU but the worst average path length. This is because their objective is energy saving primarily. However, applying NSP and NMU on top of them improves energy saving. NSP and NMU balance the trade-off between performance and energy saving. The MaxRESDN heuristics, as compared to the other heuristics, has the maximum RESDN value. By maximizing the RESDN value, the MaxRESDN has shown closer results to the best combination algorithm in terms of both energy saving and average path length. Assuming that the optimal values for *Umin* and *Umax* parameters are determined for traffic volumes, the MaxRESDN heuristics gives the maximum RESDN that achieves traffic proportional energy consumption and keeps the trade-off between energy efficiency and network performance.

Figure 10 shows the throughput of the algorithms with respect to the percentage of traffic volume that ranges from 20% to 90%. The throughput measurement is done on Mininet using the iperf command. The average bandwidth of links is set to 100 Mbps and the average flow rate is 7.79 Mbps. Results show that the NSP algorithm exhibits the highest throughput for all the traffic volumes, and is followed by NMU and MaxRESDN. This is because, for the three heuristics, the algorithms are initialized with the shortest paths. The maximum RESDN value by MaxRESDN as shown in Figure 6 for all the traffic volumes exhibits the least power consumption and also throughput close to that of NSP and NMU. For all the algorithms, the throughput decreases slightly as the traffic volume increases beyond 30%. Under conditions where energy saving is not applied, the throughput is expected to increase until the traffic volume is near 100% then decreases because of congestion. However, in energy saving routing algorithms, the attempt is to minimize the number of active links and switches. Therefore, the average throughput would decrease even when the traffic volume is not high (e.g., near 50%). The trend of throughput in energy saving algorithms is different than non-energy saving heuristics that focus on network performance.

Figure 11 shows the delays measured in milliseconds with respect to traffic volume ranging from 20% to 90%. The NSP algorithm exhibits the lowest delay followed by NMU. This is because both NSP and NMU are initialized by the shortest

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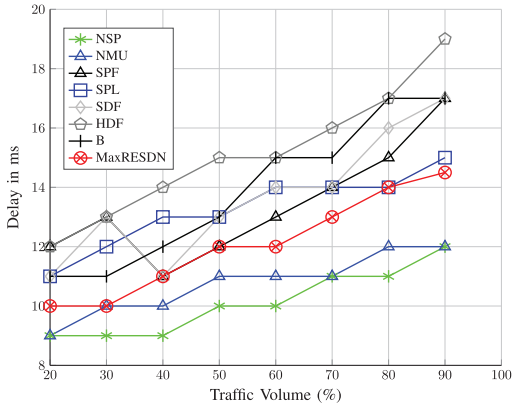

Fig. 11. Delay in milliseconds versus traffic volume ranging from 20% to 90%.

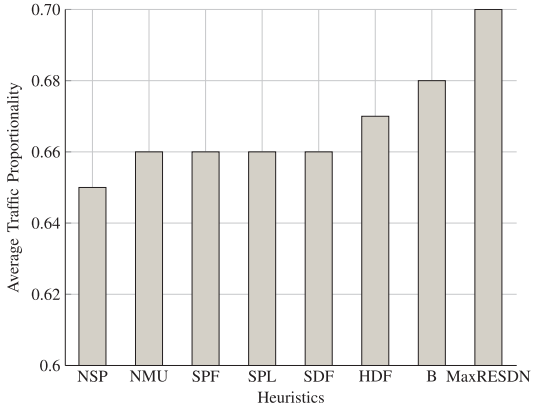

Fig. 13. Average traffic proportionality of MaxRESDN as compared to other heuristics.

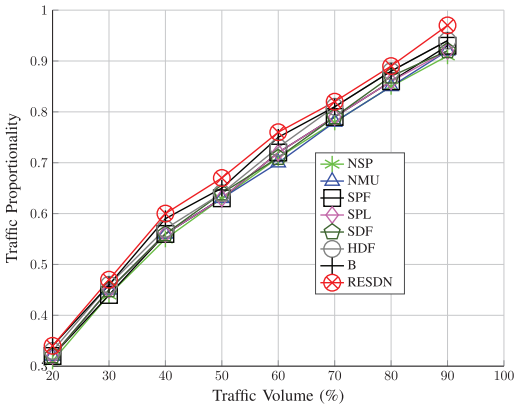

Fig. 12. Traffic proportionality versus traffic volume ranging from 20% to 90%.

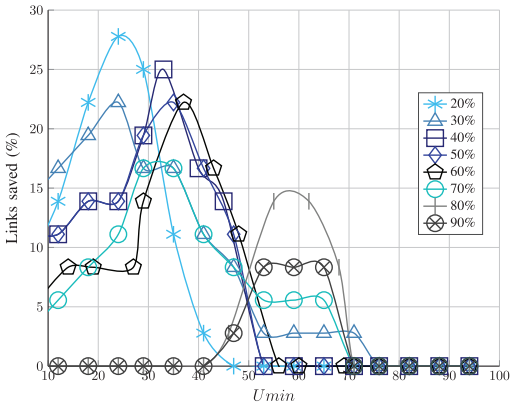

Fig. 14. Effect of utility parameter *Umin* values on the percentage of links saved for different traffic volumes ranging from 20% to 90% by fixing *Umax* at 95%.

paths. The SDF algorithm is 3 to 5 milliseconds worse in delay than the NSP algorithm. The MaxRESDN algorithm is 1 to 3 milliseconds close to the NSP algorithm. It can be observed that for all of the energy saving heuristics, the delay tends to slightly increase with the increasing traffic volume. This is because all the energy saving algorithms attempt to minimize the number of links and switches used.

Figure 12 shows the traffic proportionality of heuristics for traffic volume ranging from 20% to 90%. The MaxRESDN heuristics have demonstrated the maximum traffic proportionality value. The B heuristics, which is an attempt of keeping the trade-off between performance and energy saving, has a closer traffic proportionality to our approach.

Figure 13 shows average traffic proportionality of all traffic volumes for MaxRESDN algorithm in comparison to the other algorithms (in Table V). The MaxRESDN heuristics exhibits the largest traffic proportionality in terms of link energy consumption. NSP algorithm has the lowest traffic proportionality. As compared to those heuristics which give priority to network performance, such as NSP and SPF, our approach MaxRESDN is 4 to 5% better in traffic proportionality. It is also 3 to 4% better in traffic proportionality in comparison to the heuristics that give priority to energy saving and maximizing utility such as NSP and HDF.

D. Analysis of Utility Parameters

A significant feature of the MaxRESDN algorithm is its dependence on the link utility interval parameters. In this subsection, the trends of link utility interval parameters *Umin* and *Umax* versus traffic volume and percentage of links saved are analyzed. For the fixed *Umax* value of 95%, we study the effect of the *Umin* parameter on the percentage of links saved for traffic volume ranging from 20% to 90% *Umin*. Likewise, for the fixed *Umin* value of 30%, we study the effect of *Umax* value on the percentage of links saved. Understanding the relations for values of *Umin* and *Umax* versus the traffic volume would be significant in predicting the utility parameters for future use.

Figure 14 shows the effect of the *Umin* parameter on the percentage of links saved for traffic volumes ranging from 20% to 90%. Fixing *Umax* at 95%, and changing *Umin* value from 10% to 90%, the results show that the value of *Umin* that maximizes the links saved increases until it reaches a peak and then drops. As the traffic volume gets larger, the peak *Umin* value increases. The findings indicate similar trend for different traffic volumes.

Figure 15 shows the effect of the *Umax* value on the percentage of links saved for traffic volumes ranging from 20% to 90% where each line shows the results for a specific traffic

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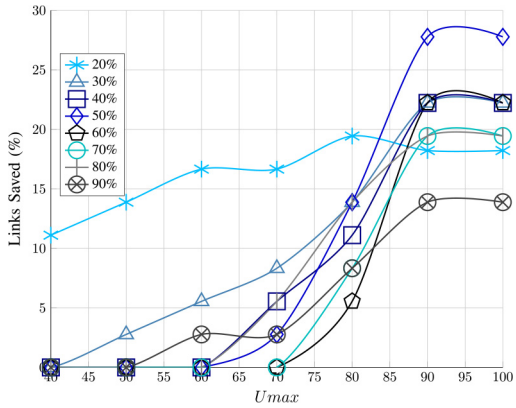

Fig. 15. Effect of utility parameter *Umax* values on the percentage of links saved for different traffic volumes ranging from 20% to 90% by fixing *Umin* at 30%.

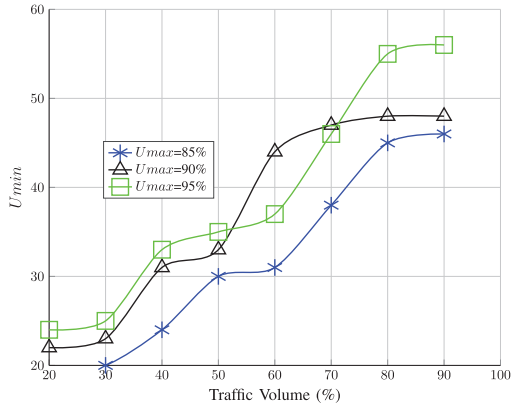

Fig. 16. The effect of the traffic volume ranging from 20% to 90% on the value of *Umin* and *Umax* that maximizes the percentage of links saved fixing *Umax* values to 85%, 90% and 95%.

volume. The *Umin* parameter is set to 30% and the *Umax* value ranges from 40% to 100%. For every flow size, the *Umax* value that maximizes the number of links saved increases until it reaches a peak and then it mainly remains constant except for the 20% traffic volume where there is a slight decrease.

Figure 16 shows the relationship between the traffic volume and the *Umin* value that maximizes the energy saving. It shows that as the traffic volume increases the *Umin* that maximizes energy saving increases. The reason for this is because the utilities of the links increase with traffic volume. For example when the percentage of traffic is 90%, the link with minimum utility is above 40%. Which means that if we pick *Umin* value less than 40%, the candidate list would be empty. Similarly as the traffic volume decrease to 20%, the link utilities starts from 12% and we will have more links in the candidate list. According to the results given in Figures 14 and 16, the problem that need to be solved hence is to find or estimate the peak *Umin* value for a given traffic volume.

Figure 17 illustrates the trend of *Umax* value in relation to traffic volume by fixing the value of *Umin* at 30%. It demonstrates that as the volume of traffic increases, *Umax* value maximizing the energy saving exhibits an increasing trend. It also shows that for traffic volume more than 72%, *Umax* value has a constant value near 95%. According to the

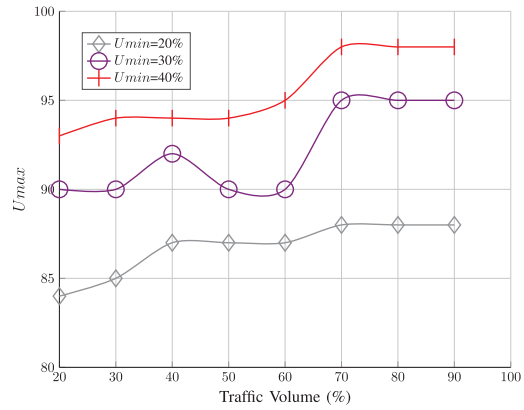

Fig. 17. The effect of the traffic volume ranging from 20% to 90% on the value of *Umax* that maximizes the percentage of links saved fixing *Umin* values to 20%, 30%, and 40%.

results depicted in Figures 17 and 16, determining the value of *Umin* and *Umax* has a direct impact on the performance of MaxRESDN algorithm.

VII. Conclusion

The flexibility of network control enabled by SDN can be used for energy efficient routing. However, minimizing energy consuming devices without compromising network performance is a challenging task. We proposed energy efficiency metric named RESDN (Ratio for Energy Saving in SDN) that puts the utility of links under consideration. Unlike other energy efficiency metrics, RESDN is based on the link utility interval parameters, and captures the dynamic network changes in SDN. We also provided an IP formulation with the objective of increasing the RESDN of a network environment and a heuristics named MaxRESDN that maximizes energy saving. The experiments are conducted on Mininet network emulator and POX controller using the GEANT network topology and dynamic traffic traces. We also simulated the power consumption of OvSwitch, NEC PF5240, and Zodiac FX OpenFlow switches.

Experimental results show that maximizing the RESDN value improves energy efficiency while maintaining acceptable network performance. In comparison to other utility-based heuristics for energy efficient routing, MaxRESDN is up to 30% better in the number of links saved and achieves up to 14.7 watts, 10 watts, and 3.2 watts less power consumption for NEC, OVS and Zodiac FX switches, respectively. By achieving the maximum RESDN value, our proposed approach performs close to the solutions that give priority to network performance in terms of average path length, throughput, and delay. It also has 3 to 5% higher traffic proportionality, and maintains the trade-off between network performance and energy efficiency. Since the performance of the MaxRESDN heuristics greatly depends on the value of utility interval parameters, we performed a detailed analysis of the parameters with regard to network traffic volume.

As future work, we aim to utilize supervised and reinforcement machine learning techniques to predict the link utility interval parameters in an efficient manner. We also

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plan to apply the idea behind RESDN metric to incorporate end systems in a data center platform, and improve scalability through parallelizing the MaxRESDN algorithm to handle multiple flow arrivals concurrently.

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